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Assignment - 5

Q1 Define Pollution & its types?

Ans

Pollution:- Pollution is the introduction of harmful substances or contaminants into the natural environment that causes adverse effects on living beings & the ecosystem.

It makes air, water, soil & surroundings impure or unsafe.

Types of Pollution :-

(1) **Air Pollution:-** Contamination of the air by harmful gases, dust, smoke or chemicals.
ex. vehicles emission, factory smoke.

(2) **Water pollution-** Contamination of water bodies like rivers, lakes & oceans by waste materials.
ex. Industrial discharge, sewage, oil spoils.

(3) **Soil pollution-** Degradation of land by dumping solid waste, chemicals & pesticides.
ex Plastic waste, fertilizers, industrial waste.

(4) **Noise Pollution-** Excessive, unwanted sound that harms health & well-being.
ex Traffic noise, loudspeakers, machinery.

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(5) Thermal Pollution:- Rise in temperature of water bodies due to industrial discharge of hot water.
ex Power plants releasing heated water into rivers.

(6) Light Pollution:- Excessive artificial light that disrupts ecosystem & human sleep.
ex Bright streetlights, billboards.

(7) Radioactive Pollution- Release of radioactive substances into the enviornment.
ex Nuclear power plants accidents, improper disposal of radioactive waste.

Q2 Explain Radioactive Pollution & how can we control radio-active waste.

Ans Radioactive Pollution is the contamination of the enviornment with Radioactive substances that emit harmful ionizing radiation (like alpha, beta & gamma rays)

This radiation can damage living cells, causes mutations, cancer and long-term enviornment harm.

Source of Radio Active Pollution-

(1) Nuclear Power Plants Accidents - Chernobyl, Fukushima.
(2) Nuclear weapon testing & explosion.
(3) Improper disposal of radioactive waste - from hospital, industries, research labs.

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(4) Mining & processing of radioactive ores - Uranium thorium.

(5) Natural sources - radon gas from rocks.

Harmful effects -

- Genetic mutations & cancer
- Damage to organs (especially bone marrow, thyroid).
- Death of plants & animals.
- Long-term soil & water contamination.

Control of Radioactive waste -

(1) Proper storage - Store radioactive waste in shield ~~and~~ Leak-proof containers (usually lead-lined steel).

(2) Deep Geological Disposal - Bury high-level waste deep underground in stable rock formations.

(3) Dilution & Dispersion - low-level waste can be diluted & discharged into water bodies under strict regulations (very controlled.)

(4) Reprocessing & Recycling - Recover usable materials (like ~~a~~ uranium & plutonium) from spent nuclear fuel to reduce waste volume.

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Q3. Define the Solid waste & the solid waste management.

Ans **Solid waste** - Solid waste refers to unwanted or discharged solid materials that are generated from human activities & are no longer useful or needed.

ex Household garbage, plastic bags, paper, food waste, broken furniture, industrial scrap, construction debris.

* **Solid waste Management** Solid waste management is the systematic process of collecting, storing, transporting, treating, and disposing of solid waste in a way that is safe for human health & the environment.

It aims to reduce the negative effects of waste, reuse resources & recycle materials whenever possible.

Main step of Solid waste management:-

~~(1) Waste management~~

(1) Waste Generation - Creation of waste from homes, industries, markets etc.

(2) Collection - Gathering waste from households, streets & industries.

(3) Storage - Storing waste in bins or containers before transportation.

(4) Transportation - Carrying waste to treatment or disposal sites.

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(5) Segregation - Seperating waste into biodegradable non - biodegradable, recyclable & hazardous categories.

(6) Treatment & processing - Recycling, composting, incineration or converting waste into energy.

(7) Final Disposal - Safe disposal in landfills or engineeried dumpsites.

Q4. Describe the sewage treatment process with the help of flow chart.

Ans Sewage treatment is the process of removing contaminants from wastewater (Sewage) coming from homes, industries & commercial plans to make it safe for the environment befor releasing it into rivers, lakes or using it for reuse.

Steps -

1. Preliminary Treatment - Removes large solids and debris.
   - screening & grit removal.

2. Primary Treatment - Settles out suspended Solids.
   - Sedimentation tanks - Solid settle as sludge, oil float as scum.

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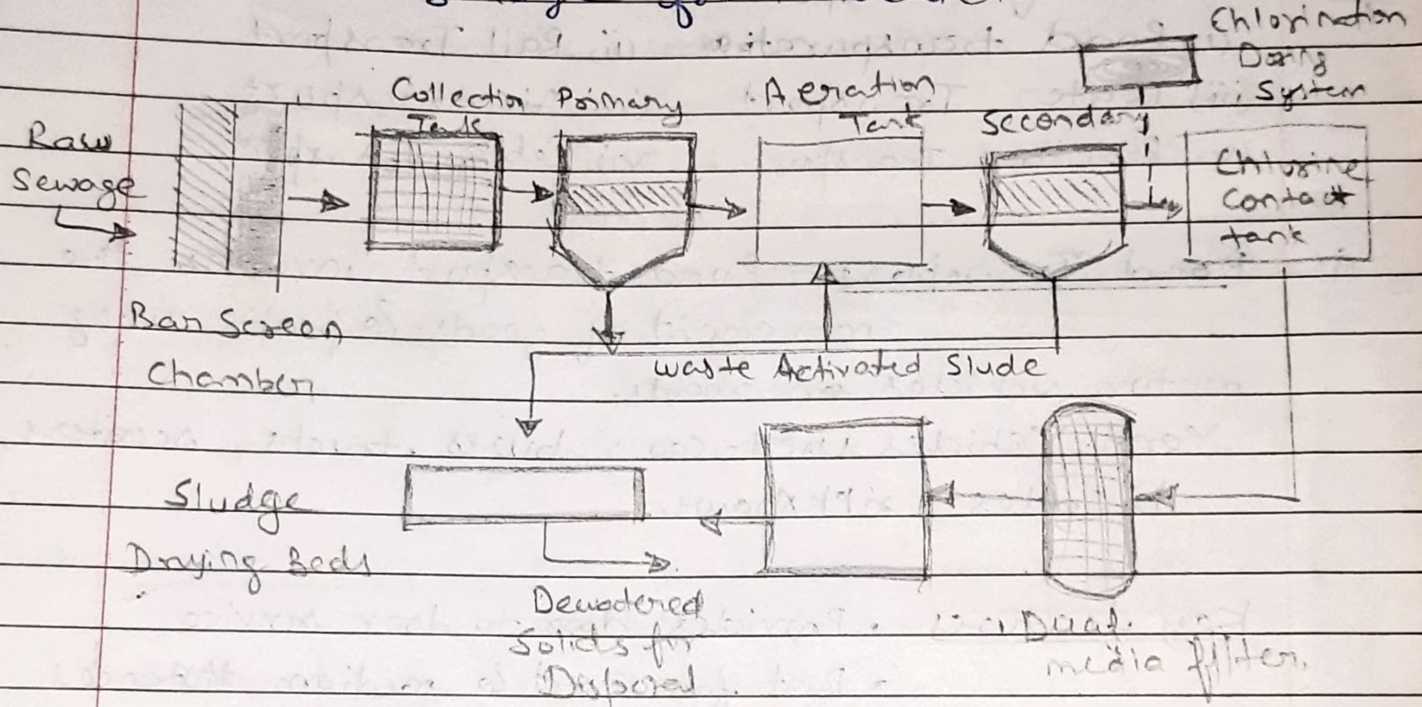
3. Secondary Treatment (Biological Treatment)
Removes dissolved & biodegradable organic matter.
Activated sludge Process (or trickling filter) microorganisms break down waste.

4. Secondary sedimentation - Removes biological sludge from treated water



Sewage Treatment Plant

5. Tertiary (Advanced) treatment (optional) - Removes nutrients, pathogens and remaining pollutants
Filtration, Disinfection (Chlorination, UV), nutrient removal.

6. Sludge Treatment & Disposal - stabilizes & disposes of sludge (can be composted, incinerated or landfilled)

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7. Treated Effluent Disposal - Safe water is discharged into rivers, lakes or used for irrigation.

Q5 What are the different modes of Transportation Explain each of one.

Ans Modes of Transportation -
(i) Road transporation (ii) Rail Transport
(iii) Water Transport (iv) Air transport.
(v) Pipeline Transport (vi) Cable Transport.

(i) Road Transport - Road transport involves the movement of goods & people using motor vehicles on roads.
Vech. Vehicles used - cars, buses, trucks, scooters, bicycles, rickshaws.

Key features
- Provides door to door service
- Best for short & medium distances
- Flexible route & timing.
- Suitable for urban & rural areas.

Advantage -
- Easy & quick to arrange.
- Cheaper for small cargo & passenger.
- Useful for reaching remote village.

Disadvantage -
- Slower over long distances.
- Roads may suffer from traffic congestion & accidents.

Example

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(2) Rail Transport - Rail transport uses trains running on railways tracks for moving passengers & heavy goods.

Key features:- • Ideal for long-distance, bulk transport.
• Operates on fixed routes & schedule.
• Requires railway station for loading / ~~upto~~ unloading.

Advantage:- • Economical for heavy & bulky goods
• Safe & reliable, with less environment pollution.
• Suitable for long journeys.

Disadvantage:- • No direct door to door service
• Limited to areas with rail infrastructure.

(3) Water Transport :- water transport uses boats, ships & ferries over rivers (inland) & seas (oceans) (marine).

Types:- Inland water transport - River, canals, lakes.
Ocean / Sea transport :- Ships & vessels for international trade.

Key features :- Best for heavy & bulky cargo.
Important for international trade.

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Advantage :- Cheapest for large quantities over long distance.
Less fuel consumption.
- Ideal for international trade. (Across continents)

Disadvantage :- • Slow compared to air & road transport.
• Affected by weather & seasonal condition
• Requires ports & harbors.

4. Air transport :- Air transport involves airplanes & helicopters flying in the sky to transport people & goods. It is the fastest means of transport & is essential for international travel & emergency services.

Types of Air transport -
Commercial airlines - Domestic & international flights.
Cargo airlines - Air freight carriers like FedEx, DHL
Charter flights & helicopters.

Advantage • Very fast over long distance.
• Can reach remote & inaccessible area quickly.
• Suitable for high value & perishable goods

Disadvantage -
• High cost of transport.
• Affected by weather conditions (fog, storms)
• Limited capacity for heavy or bulky goods.

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5. Pipe line Transportation Pipline are used to transport liquids & gases over long distances. This includes petroleum products, water & natural gas.

Types liquid – Cruide oil, petroleum.
Gas – Natural gas, liquified gas (LPG)
Slurry – Transport of ore mixed with water.

Advantage – Continious efficient, safe, minimal enviornmental impact compared to road.
• No traffic congestion.

Limitation – limited to specific types of cargo
• High installation costs
• Difficult to detect & repair leaks.

6. Cable Transportation – Cable transport systems such as cable cars, gondolas, funiculars & ropeways. These are often used in mountainous regions or for tourism.

Type Aerial cableways – Ropeways, gondolas.
Surface cable cars – Funicular railways.

Advantage
• Ideal for hilly & mountainous area.
• Provides scenic travel experiences
• Less land required for infrastructure

Disadvantage –
– Limited carrying capacity.
• High construction and maintenence cost
• Not suitable for long distance.